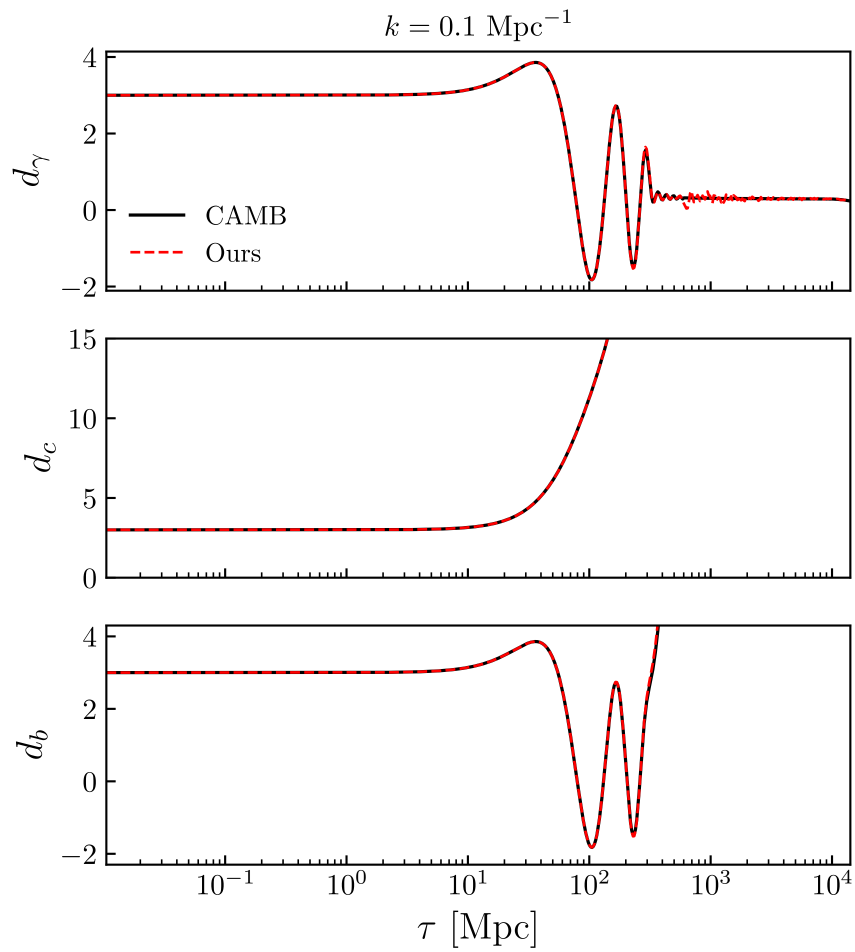
Boltzmann Code - Perturbations

Last week we evolved a single Fourier mode of a radiation-only universe. This week we will extend the same machinery to the full ΛCDM perturbation system. In particular, we'll add in CDM and baryons and couple the photons and baryons through Thomson scattering. By the end of this lab you will have a servicible Einstein-Boltzmann solver.



Setup: We continue to work in conformal Newtonian gauge with metric

$$ds^2 = a^2(\tau)\left[-(1+2\Phi)\,d\tau^2 + (1-2\Psi)\,d\vec{r}^{\,2}\right],$$

track the evolution of the perturbation $d_a \equiv \delta_a/(1+w_a) - 3\Psi$ for each species, together with the velocity potential u_a, anisotropic-stress potential π_a, and the higher-l multipole tower $d_{l,a}$ for collisionless species. Remember that $d_{0,\nu} = d_\nu$, $d_{1,\nu} = u_\nu$, $d_{2,\nu} = (3/2)\pi_\nu$. The same multipole structure we had only for neutrinos last time will now also apply to photons after we leave the tight-coupling regime.

The new species are **CDM** and **baryons**. In the regime of small perturbations, we only need to track (d_c, u_c, d_b, u_b). Instead of radiation domination like we had in `nu_phase_shift.pdf`, we will use our full Planck ΛCDM background cosmology which we implemented in `background.pdf`, with the ionization history $x_e(z)$ from `recombination.pdf` and `reionization.pdf`.

The metric perturbations are still determined at every time step directly from species perturbations (now with a ranging over all four species):

$$\boxed{\Psi = -\frac{\gamma_{\rm tot}\,(d_{\rm tot} + 3\mathcal{H}\,u_{\rm tot})}{k^2 + 3\gamma_{\rm tot}}, \qquad \Phi = \Psi - 3\gamma_{\rm tot}\,\pi_{\rm tot},} \tag{1}$$

where $\gamma_a \equiv 4\pi G a^2(\bar\rho_a + \bar P_a)$, $\gamma_{\rm tot} = \sum_a \gamma_a$, and the totals are weighted averages

$$d_{\rm tot} = \sum_a x_a\,d_a, \quad u_{\rm tot} = \sum_a x_a\,u_a, \quad \pi_{\rm tot} = \sum_a x_a\,\pi_a, \qquad x_a \equiv \frac{\gamma_a}{\gamma_{\rm tot}}.$$

For CDM and baryons we have $\bar P_a = 0$ so $\gamma_a = 4\pi G a^2 \bar\rho_a$.

> **Part 1:** Extend last week's `metric_perturbations` to take all four species as input and return (Φ, Ψ) using Eq. (1).

State vector: Following the same conventions as last week, the full state vector is

$$y = (\underbrace{d_c, u_c}_{\text{CDM}}, \underbrace{d_b, u_b}_{\text{baryon}}, \underbrace{d_\gamma, u_\gamma, \pi_\gamma, d_{3,\gamma}, \ldots, d_{\texttt{LMAX}_\gamma,\gamma}}_{\text{photon}}, \underbrace{d_\nu, u_\nu, \pi_\nu, d_{3,\nu}, \ldots, d_{\texttt{LMAX}_\nu,\nu}}_{\text{neutrino}}).$$

We take $\texttt{LMAX}_\gamma = \texttt{LMAX}_\nu = 15$ for this lab.

Initial conditions: We start integration deep in the radiation era, when the mode is still frozen on superhorizon scales. The initial condition for adiabatic growing modes are the same as last week:

$$\boxed{d_a = -3\zeta, \qquad u_a = \frac{5\tau_{\text{init}}\,\zeta}{15 + 4R_\nu}, \qquad \pi_\nu = \frac{2\tau_{\text{init}}^2\,\zeta}{3(15 + 4R_\nu)},} \tag{2}$$

with all higher-l neutrino multipoles vanishingly small. We set $\zeta = -1$ to match convention of the transfer functions that are easy to pull out of `CAMB`.

> **Part 2:** Write `initial_state(tau_init, k, bg)` that returns the initial condition for the full state vector using Eq. (2). This is essentially last week's `initial_state` with two extra blocks for CDM and baryons.

Tight-coupling approximation: Before recombination, the rate at which CMB photons Thomson-scatter off free electrons is very fast. So, the Thomson scattering terms in the photon and baryon Euler equations,

$$u'_\gamma \supsetsim \sigma_T x_e\,(u_b - u_\gamma), \qquad u'_b \supsetsim (1/R_b)\,\sigma_T x_e\,(u_\gamma - u_b), \qquad R_b \equiv \frac{3\bar\rho_b}{4\bar\rho_\gamma},$$

are huge relative to other relevant time scales. This gives us a nasty stiff system. To get around this, we can use the tight-coupling approximation (TCA) where we evolve the combined photon-baryon fluid as a single entity while keeping a few low-order corrections in the photon mean-free path $\tau_c \equiv 1/(\sigma_T a n_e)$.
The TCA equations for the photon-baryon fluid are

$$\boxed{d'_\gamma = -k^2 u_\gamma, \qquad u'_\gamma = -\mathcal{F}\,u_\gamma \;+\; \frac{d_\gamma}{3(1+R_b)} \;+\; \Phi + \frac{\Psi}{1+R_b},} \tag{3}$$

where the friction coefficient

$$\boxed{\mathcal{F} \equiv \frac{\mathcal{H} R_b}{1+R_b} + 2k^2\tau_d, \qquad \tau_d \equiv \frac{\tau_c}{6}\left[1 - \frac{14}{15(1+R_b)} + \frac{1}{(1+R_b)^2}\right],} \tag{4}$$

contains two physically distinct pieces. The $\mathcal{H}R_b/(1+R_b)$ term is the friction from the baryon mass-loading on the photon fluid: in the limit $R_b \to 0$ (no baryons) it vanishes and the photons oscillate freely with sound speed $c_s^2 = 1/3$. For nonzero R_b the baryons drag, the sound speed slows to $c_s^2 = 1/(3(1+R_b))$, and the fluid loses kinetic energy to expansion. The $2k^2\tau_d$ piece is **Silk damping**, which quantifies the small slip between photons and baryons from photon diffusion. See `lecture_notes/section_06.pdf` for more discussion.

The baryons are locked to the photons,

$$\boxed{d_b' = -k^2 u_\gamma, \qquad u_b' = u_\gamma'.} \tag{5}$$

CDM is collisionless throughout, so its equations are the same in TCA and the full system:

$$\boxed{d_c' = -k^2 u_c, \qquad u_c' = -\mathcal{H}\, u_c + \Phi.} \tag{6}$$

Note that CDM is non-relativistic, so only Φ drives the Euler equation. You can put this in contrast with the radiation Euler equation where both Φ and Ψ appear as sources.

The massless neutrinos are also collisionless and are evolved using exactly the same equations as last week. Reproducing them here for completeness: the monopole/dipole/quadrupole are

$$\boxed{d_\nu' = -k^2 u_\nu, \qquad u_\nu' = \tfrac{1}{3} d_\nu - k^2 \pi_\nu + \Phi + \Psi, \qquad \pi_\nu' = \tfrac{4}{15} u_\nu - \tfrac{2}{5} k^2 d_{3,\nu},} \tag{7}$$

the higher multipoles ($l \geq 3$) obey the recursion

$$\boxed{d_{l,\nu}' = \frac{l}{2l+1}\, d_{l-1,\nu} \;-\; \frac{l+1}{2l+1}\, k^2\, d_{l+1,\nu}, \qquad l \geq 3,} \tag{8}$$

with $d_{2,\nu} = (3/2)\pi_\nu$ at $l = 3$, and the streaming closure at $l = \mathtt{LMAX}_\nu$ is

$$\boxed{d_{\mathtt{LMAX}_\nu,\nu}' = d_{\mathtt{LMAX}_\nu-1,\nu} - \frac{\mathtt{LMAX}_\nu + 1}{\tau}\, d_{\mathtt{LMAX}_\nu,\nu}.} \tag{9}$$

> **Part 3:** Write **tca_rhs(tau, y, k, bg)** that assembles the time derivatives in TCA using Eqs. (3), (5), (6), and the neutrino block Eqs. (7), (8), (9). You will need helpers **scale_factor_from_tau(tau)** and a Thomson-rate $\sigma_T a n_e$ as a function of τ. With this, you can now integrate the TCA system with **scipy.integrate.solve_ivp** starting from $\tau_{\rm init} = 10^{-3}/k$.

Validation – TCA: Before moving on, let's see how the TCA solution holds up against **CAMB**. We will compare against **CAMB**'s Newtonian-gauge perturbations using **camb.symbolic** to access δ_γ, q_γ, δ_c, δ_b, δ_ν, and Φ_N. Note that Φ_N from CAMB is our Ψ. Translating to our variables,

$$d_\gamma = \tfrac{3}{4}\delta_\gamma - 3\Psi, \quad u_\gamma = \tfrac{3}{4}q_\gamma/k, \quad d_c = \delta_c - 3\Psi, \quad d_b = \delta_b - 3\Psi, \quad d_\nu = \tfrac{3}{4}\delta_\nu - 3\Psi.$$

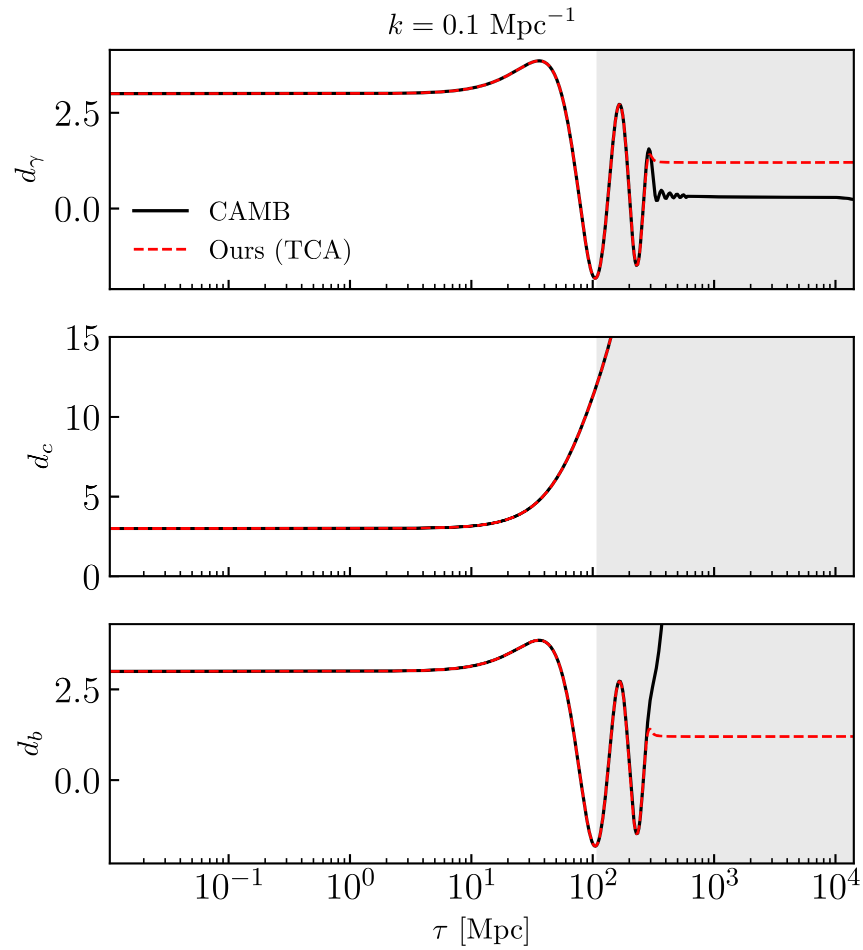
In code this look roughly like

```python
import camb
from camb.symbolic import Delta_g, Delta_c, Delta_b, Delta_r, q_g, Phi_N

pars = camb.set_params(...)
results = camb.get_results(pars)


def camb_perts(k, tau):
    out = results.get_time_evolution(
        k, tau,
        vars=[Delta_g, q_g, Delta_c, Delta_b, Delta_r, Phi_N],
        frame='Newtonian',
    )
    delta_g, q_g_c, delta_c, delta_b, delta_r, phi_n = out.T
    return {
        'd_g': 0.75 * delta_g - 3.0 * phi_n,
        'u_g': 0.75 * q_g_c / k,
        'd_c': delta_c        - 3.0 * phi_n,
        'd_b': delta_b        - 3.0 * phi_n,
        'd_n': 0.75 * delta_r - 3.0 * phi_n,
    }
```

For $k = 0.1\,\mathrm{Mpc}^{-1}$ you should get something like



Our d_γ, d_c, and d_b agree with `CAMB` well at early times, but diverge once we pass into the regime where the tight-coupling approximation fails (the grey shaded region). This regime we define as $\tau > \tau_{\text{switch}}$ with $k\tau_c(\tau_{\text{switch}}) \simeq 0.02$. Basically, we say that once a photon's mean-free-path approaches a tenth of a wavelength, the photon-baryon plasma can no longer be treated as a single fluid.

Full system: Once $\tau > \tau_{\text{switch}}$, we keep the Thomson rate $\sigma_T a n_e$ explicitly in the equations rather than expanding around $\sigma_T a n_e \to \infty$. We'll now track Photons and baryon perturbations separately.

Baryons. The baryon continuity equation is the same as before. The Euler equation gains a Thomson drag from the photons:

$$\boxed{d_b' = -k^2 u_b, \qquad u_b' = -\mathcal{H}\, u_b + \Phi + \frac{\sigma_T a n_e}{R_b}\,(u_\gamma - u_b).} \tag{10}$$

The factor of $1/R_b$ comes from momentum conservation, the momentum transferred to baryons is the momentum lost by photons.

Photons. The photon continuity, Euler, and quadrupole equations are

$$\boxed{\begin{aligned} d_\gamma' &= -k^2 u_\gamma, \\ u_\gamma' &= \tfrac{1}{3} d_\gamma \;-\; k^2 \pi_\gamma \;+\; \Phi + \Psi \;+\; \sigma_T a n_e\,(u_b - u_\gamma), \\ \pi_\gamma' &= \tfrac{4}{15} u_\gamma \;-\; \tfrac{2}{5} k^2 d_{3,\gamma} \;-\; \sigma_T a n_e\, \pi_\gamma. \end{aligned}} \tag{11}$$

Two changes from the neutrino case: (1) the Thomson term $\sigma_T a n_e\,(u_b - u_\gamma)$ in the Euler equation and (2) the Thomson damping $-\sigma_T a n_e\, \pi_\gamma$ in the quadrupole equation (this is because Thomson scattering is isotropic in the rest frame of the electron, so it tries to isotropizes the photon distribution by damping all $l \geq 2$ moments).

The higher-l photon multipoles obey the same recursion as neutrinos but with an additional Thomson damping term that kicks in for all $l \geq 3$:

$$\boxed{d'_{l,\gamma} = \frac{l}{2l+1}\,d_{l-1,\gamma} - \frac{l+1}{2l+1}\,k^2\,d_{l+1,\gamma} - \sigma_T a n_e\,d_{l,\gamma}, \qquad l \geq 3,} \tag{12}$$

with $d_{2,\gamma} = (3/2)\pi_\gamma$ used at $l = 3$. The streaming closure at $l = \texttt{LMAX}_\gamma$ inherits the same Thomson damping,

$$\boxed{d'_{\texttt{LMAX}_\gamma,\gamma} = d_{\texttt{LMAX}_\gamma-1,\gamma} - \frac{\texttt{LMAX}_\gamma + 1}{\tau}\,d_{\texttt{LMAX}_\gamma,\gamma} - \sigma_T a n_e\,d_{\texttt{LMAX}_\gamma,\gamma}.} \tag{13}$$

The CDM and neutrino equations are unchanged from TCA.

Part 4: Write **`full_rhs(tau, y, k, bg)`** that assembles the time derivatives for the full system using Eqs. (10), (11), (12), (13), and the unchanged CDM and neutrino blocks. Then write **`solve_perturbations(k)`** that (i) finds $\tau_{\rm switch}$ from $k\tau_c \simeq 0.02$, (ii) integrates **`tca_rhs`** from $\tau_{\rm init}$ to $\tau_{\rm switch}$, and (iii) hands the final TCA state to **`full_rhs`** and integrates it the rest of the way to today.

Validation – full system: Solve the full system for a bunch of different k. For $k = 0.1\,\text{Mpc}^{-1}$, you should reproduce the plot on the first page.

References:

- **Bashinsky & Seljak (2003)**, *Signatures of relativistic neutrinos in CMB anisotropy and matter clustering*, astro-ph/0310198.
  Like last week, this is the paper whose notation we're following.

- **Ma & Bertschinger (1995)**, *Cosmological Perturbation Theory in the Synchronous and Conformal Newtonian Gauges*, astro-ph/9506072.
  The standard reference for linear perturbations.